

OC Core 1.3 / Biology Domain Packet

Status Snapshot

- Current state: `VALIDATED_ANCHOR_ACTIVE`
- Target state: `VALIDATED_ANCHOR_ACTIVE`
- Scientific class: `THEOREM_NATIVE`
- Trace status: `TRACE_COMPLETE`
- Closure verdict: `PASS`
- Quantitative pass result: `PASS`
- Next required action: `MAINTENANCE_ONLY`
- Replay status: `PASS_REPLAYABLE`

Theorem-to-Observable Map

- K4/K5/K6/K7 organizational thresholds lawfully generate the bounded biology packet over expression, state-transition, and response-onset observable families.
- The promoted biology packet binds source-native biological organization claims to pinned assay observables without expanding to unrestricted biological universalization.
- Every promoted biological benchmark family is traced through the locked K4/K5/K6/K7 route, open assay datasets, and held-out replay.

Benchmark Dataset Manifest

- `BIO_BENCH_001`: `expression_signature_residual` | NCBI GEO | <https://www.ncbi.nlm.nih.gov/geo/> | held-out: reserve one time-series family for held-out replay
- `BIO_BENCH_002`: `state_transition_order_error` | NCBI GEO | <https://www.ncbi.nlm.nih.gov/geo/> | held-out: reserve one cell-state transition family
- `BIO_BENCH_003`: `response_onset_residual` | official open biological assay repositories | <https://www.ncbi.nlm.nih.gov/geo/> | held-out: reserve one response-signature family

Pinned Official Route Protocol

- `BIO_ROUTE_001`: NCBI GEO (`OFFICIAL_OPEN_DATA_HUB`) -> <https://www.ncbi.nlm.nih.gov/geo/> | purpose: Open assay and expression benchmark families

Measurable Outputs

- `expression_signature_residual`
- `state_transition_order_error`
- `response_onset_residual`

Measurement Schema

Choose one bounded biological lane and evaluate predicted state transitions or response signatures against open assay data.

Acceptance Criterion

The promoted biological packet reproduces the declared measurable signatures on held-out datasets without widening the claim boundary.

Falsifier

If the bounded signature family fails on held-out datasets, the biological packet remains frontier-bound.

Quantitative Acceptance Thresholds

- coverage_ratio_required: 1.0
- expected_calibration_error_max: 0.05
- normalized_error_max_max_sigma: 5.0
- normalized_error_mean_abs_max_sigma: 1.0
- normalized_error_p95_max_sigma: 2.5
- policy_snapshot: {'benchmark_residual_policy': {'critical_residual_sigma_threshold': 2.5, 'schema_id': 'LOGION_BENCHMARK_RESIDUAL_POLICY_v1', 'severe_residual_sigma_threshold': 1.5}, 'prediction_accuracy_policy': {'brier_score_max': 0.08, 'cases_total_required': 30, 'coverage_ratio_required': 1.0, 'critical_failure_f1_min': 0.87, 'critical_failure_precision_min': 0.85, 'critical_failure_recall_min': 0.9, 'expected_calibration_error_max': 0.05, 'schema_id': 'LOGION_PREDICTION_ACCURACY_POLICY_v2'}, 'prediction_sigma_policy': {'cases_total_min': 30, 'normalized_error_max_max_sigma': 5.0, 'normalized_error_mean_abs_max_sigma': 1.0, 'normalized_error_p95_max_sigma': 2.5, 'schema_id': 'LOGION_PREDICTION_SIGMA_POLICY_v1', 'tail_breach_count_max': 0}}
- tail_breach_count_max: 0

Replay Harness

- Replay command: validation/run_all.py --qa-only (public replay umbrella; domain: BIOLOGY)
- Evidence bar: HYBRID_ESCALATION
- Protocol id: OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES

Institute-run Escalation

- Current escalation status: NOT_REQUIRED_UNLESS_NEW_OBSERVABLE_FAMILY_ENTERS
- Escalation trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_SIGNATURE_CLASS_UNDER
- Measurement wave id: INSTITUTE_RUN::BIOLOGY::WAVE_3A
- Measurement plan: Run a bounded institute observation or assay campaign only for the explicitly declared biological lane; no widening of biological scope is allowed.

Blocking IDs

- None.